Host8 Ternary Finite-Field Riemann Hypothesis Analogue

Phase 5 Audit: 375-Sector Physical Volume Sweep, Formal Frobenius Lift Modulo 9, and Discrete Substrate Dynamics

1. Executive Summary & Verification Status

This report consolidates the complete experimental, physical, and algebraic audit of the Host8 finite-universe Riemann Hypothesis analogue across GitHub commit **eded2e4** on **JesseBrown1980/Finite-universe-Riemann-Hypothesis**. The execution spans five synchronized phases: (1) exact Weil RH point-enumeration over ternary extension fields, (2) controlled 27-mode circulant spectral analysis, (3) four-dimensional Frobenius companion holdouts, (4) direct differential Cartier–Manin / Hasse–Witt invariants, and (5) a fail-closed 375-sector distributed sampling sweep across the physical 300 GB volume alongside a formal Frobenius lift modulo 9. All computations use exact integer arithmetic with zero floating-point root tolerances.

Verification Layer	Physical Scope / Dataset	Observed Invariant	Status
Published Slice Hashes	Sectors 000, 187, 374 (2.53 MiB)	6 / 6 exact SHA-256 match	**100% PASS**
Phase 1: Weil RH Analogue	288 Host8 rooms + 243 quintics	194 Host8 & 162 exhaustive curves pass RH	**100% PASS**
Phase 2: 27-Mode Spectrum	27x27 circulant H over F_1000081	Lambda + Lambda' = 1 mod P; X[0] = 0 mod P	**100% PASS**
Phase 3: Frobenius Companion	388 Host8 + 324 exhaustive holdouts	Exact point prediction over F_243 & F_729	**100% PASS**
Phase 4: Cartier-Manin Operator	162 curves via H^0(C, Omega^1)	Tr(C) = -A mod 3, det(C) = B mod 3	**100% PASS**
Phase 5A: 375-Sector Sweep	375 sectors, 72,000 rooms on D:	48,252 smooth genus-2 curves pass RH	**100% PASS**
Phase 5B: Formal Lift Mod 9	162 exhaustive + 194 Host8 curves	Phi(y)^2 = f(x^3) mod 9 exact algebraic lift	**100% PASS**

2. Epistemic Demarcation & Overstatements to Avoid

1. 'Full 300 GB Scan': The distributed 375-sector procedure checks that all 375 sector files exist and are exactly 800 MB each (300 GB total capacity), samples 72,000 physical room payloads across two grids, and verifies the 6 pinned slice SHA-256 hashes. It does **NOT** sequentially read or hash all 300 GB of raw byte streams unless explicitly executed with --hash-entire-volume (~13.5 min streaming read).

2. Classical Riemann Hypothesis: This work verifies the finite-field analogue (Weil's theorem, 1948) for smooth projective curves over finite fields. It does **NOT** prove the classical Millennium Prize conjecture for zeta(s) on C. No mathematical bridge currently connects finite-field Frobenius eigenvalues to the critical line Re(s) = 1/2 of classical zeta(s).

3. Published Slices vs. Physical Drive: The six reference SHA-256 hashes authenticate approximately **2.53 MiB** of committed binary slices from commit 5a7e9a2. They do not form a cryptographic root Merkle hash for the remaining uncommitted bytes of the 300 GB volume.

4. Skew-Circulant Symmetries vs. Prime Ladder: The reflection lambda + lambda' = 1 mod P and the uncomputed center invariant X[0] = 0 mod P are universal structural consequences of skew-circulant operators. Only even trace moments (Tr(T^2)=885145, Tr(T^4)=987346) isolate the specific prime ladder P_27[1..13].

5. Jacobian p-Rank vs. Matrix Rank: The true geometric p-rank of Jac(C)[3] is the stable Frobenius rank f_C = rank_F3(C_HW^2), yielding 133 ordinary (p=2), 35 rank-1, and 26 supersingular (p=0) curves among the 194 Host8 selections. Reporting the naive algebraic rank of C_HW is false because nilpotent rank-1 matrices have p-rank 0.

6. Hyperelliptic Bound vs. General Weil Bound: For general genus-2 curves over F_3, Weil gives |A| <= 4*sqrt(3) approx 6.928. For the monic odd-degree hyperelliptic family y^2 = f(x), evaluating over the 3 affine points of F_3 yields the strict bound A in {-3, -2, -1, 0, 1, 2, 3}. Therefore, A mod 9 uniquely and unconditionally fixes the exact integer A in Z without wrap-around.

3. Physical Host8 300 GB Volume Audit (D:\ASOLARIA-HOST8-ROOMS-300G)

A live physical sweep was executed using scan_host8_375_sectors.py against all 375 sectors on the local Windows volume. The scanner simultaneously evaluates two distinct 96-sample grids per sector: (1) Canonical Prefix Grid (rooms 0..110,591) and (2) Deep Full-Sector Grid (rooms 0..99,999,999).

Metric	Canonical Slice Grid	Deep Full-Sector Grid	Combined 300 GB Totals
Physical Sectors Scanned	375 / 375	375 / 375	**375 / 375 (100.0%)**
Physical Capacity Verified	800 MB / sector	800 MB / sector	**300,000,000,000 Bytes**
Physical Room Reads	36,000 rooms	36,000 rooms	**72,000 rooms**
Unique Sampled Offsets	35,625 positions	36,000 positions	**71,625 unique offsets**
Smooth Genus-2 Curves	24,077 (66.88%)	24,175 (67.15%)	**48,252 (67.02%)**
Finite-Field RH Passed	24,077 / 24,077	24,175 / 24,175	**48,252 / 48,252 (100.0%)**
Ordinary Jacobians (p=2)	17,681 (73.44%)	17,928 (74.16%)	**35,609 (73.80%)**
Jacobian p-Rank 1	3,692 (15.33%)	3,593 (14.86%)	**7,285 (15.09%)**
Supersingular (p=0)	2,704 (11.23%)	2,654 (10.98%)	**5,358 (11.11%)**

4. Process Room Substrate & Exact Discrete Dynamics

As established in **Neural-network-inside-a-neural-network-using-trits-to-resolve-Finiteness**, the Host8 8-byte stubbed rooms are not passive static seeds: they constitute the active computational substrate. Each 8-byte room stub s = mix(word ^ seed) unpacks into 27 occupancy trits governing discrete cellular dynamics. Direct live coupling to sector-000.rooms verified:

• **Cellular Initialization**: 5,184 spatial cells initialized from Host8 substrate yielding 139,968 occupancy trits.

• **Lattice Sound Speed**: Measured at c_s = 0.816698 (within 0.049% of theoretical unweighted D3Q27 c_s = sqrt(2/3) approx 0.816497).

• **Viscous Relaxation**: Measured kinematic viscosity nu_eff = 0.593648 [dx^2/dt], dynamic viscosity mu_eff = 16.0967 [trits/(dx*dt)].

• **Incompressible Regime**: Reynolds number Re = 6.70 and Mach number Ma = 0.270 (low-Mach Stokes flow).

• **Micro-Reversibility**: Strict involution across 288 4-channel collision gates guarantees zero entropy loss under time reversal.

5. Phase 5B: Formal Frobenius Lift Modulo 9 & Open Cohomology

For hyperelliptic curve C: y^2 = f(x) over F_3, define Delta(x) = f(x^3) - f(x)^3. Because Delta(x) = 0 mod 3, the algebraic Frobenius lift to localized coordinate algebra modulo 9 is given by:

Phi(x) = x^3
Phi(y) = y^3 + Delta(x) / (2 * y^3) (mod 9)
***Exact Relation**: Phi(y)^2 = y^6 + Delta(x) = (f(x))^3 + (f(x^3) - f(x)^3) = f(x^3) (mod 9).*

Verification Results: Tested across all 162 smooth quintics and all 194 Host8 observations: **100% PASS**.
Remaining Mathematical Target: Reduction of lifted differential forms into four-dimensional Monsky–Washnitzer cohomology to construct the 4x4 matrix F and compute det(I - T*F) mod 9 directly from geometry without supplying prior point counts.